

ELEC4631 Solutions Tutorial 1

1. (a) $c_A(\lambda) = \det(\lambda I - A) = (\lambda - 4)(\lambda - 2)$. Two positive eigenvalues 2 and 4, matrix is positive definite.
- (b) $c_A(\lambda) = \det(\lambda I - A) = \lambda^2 - 8\lambda - 1$. One positive and one negative eigenvalue $4 + \sqrt{17}$ and $4 - \sqrt{17}$, respectively. Matrix indefinite (not definite).
- (c) $c_A(\lambda) = \det(\lambda I - A) = (\lambda + 1)(\lambda + 3)$. Two negative eigenvalues -1 and -3 , respectively. Matrix negative definite.
2. $c_A(\lambda) = (\lambda + 3)(\lambda^2 - 6\lambda + 10) = (\lambda + 3)(\lambda - 3 - j)(\lambda - 3 + j)$. Eigenvalues are $\lambda_1 = -3$, $\lambda_2 = 3 + j$, and $\lambda_3 = 3 - j$. Eigenvectors solving $(A - \lambda_i I)v_i = 0$ for $i = 1, 2, 3$ can be chosen to be $v_1 = (1, 0, 0)^T$, $v_2 = (5, 6 + j, 5 + 7j)^T$, and $v_3 = (5, 6 - j, 5 - 7j)^T$. So,

$$V = \begin{bmatrix} 1 & 5 & 5 \\ 0 & 6 + j & 6 - j \\ 0 & 5 + 7j & 5 - 7j \end{bmatrix}$$

and $V^{-1}AV$ is diagonal.

3. (a) $c_A(\lambda) = \det(\lambda I - A) = (\lambda - 2)(\lambda - 3)^2$, eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = 3$.
- (b) Algebraic multiplicities: $\mathbf{a}_{\lambda_1} = 1$, $\mathbf{a}_{\lambda_2} = 2$.
- (c) Geometric multiplicities: $\mathbf{g}_{\lambda_1} = 1$, $\text{null}(A - \lambda_2 I) = \text{span}\{(0, 1, 0)^T\}$ so $\mathbf{g}_{\lambda_2} = 1$.
- (d) $\mathbf{g}_{\lambda_2} \neq \mathbf{a}_{\lambda_2}$, matrix not diagonalisable.
4. e_1 and e_2 can be expressed as $e_1 = f_1 + f_2$ and $e_2 = f_1 + 2f_2$, so

$$V = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix},$$

and

$$A' = VAV^{-1} = \begin{bmatrix} 7 & -5 \\ 18 & -13 \end{bmatrix}.$$